Finite-Cutoff Raw Descendant Data on the UOM Receiver Head: SK Bicomplex, Anomalous First Descent, and Conditional Hodge Normalization

Abstract

This note studies raw first-descendant data at finite cutoff on a bounded receiver-head slab using only UOM-side finite-cutoff inputs. On the slab-supported sufficiently local complex it constructs the geometric two-time bicomplex, proves the t-Ward identity, identifies the fixed-scheme physical radial Ward operator with the geometric u-derivative on compact head smearings, and derives the exact anomalous descent

$$Q_u^{\mathrm{phys}}\mathcal{O}^{(0)} + Q_t\Xi^{(1)} = \mathcal{A}^{(1)},$$

and proves that a nonzero RG slope forces a nontrivial degree-one cohomology class.

A second, conditional layer treats Hodge/Moore–Penrose normalization of alternative representatives on a selected admissible Hilbertized sector. Finally, choosing the cumulative anomaly-floor monotone yields an explicit finite-cutoff realization of the $\mathbf{2T}_{\mathrm{RG}}$ data on the slab-supported monotone sector, together with a double-cone localization statement for the canonical representatives.

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1 Scope and UOM Input Package

All external premises used below are UOM-side finite-cutoff statements; no result from the Main manuscript lane is used as a premise.

Receiver-head localization. Fix one accepted receiver cutoff $\Lambda_k < \infty$ and one bounded receiver-head time slab

$$I_k = (t_k^-, t_k^+) \subset \mathbb{R}, \qquad \mathscr{X}_k := I_k \times \Sigma_{\mathrm{rec}}.$$

All sufficiently local functionals below are supported in $\mathscr{X}_k$ and evaluated in the finite-cutoff UOM background at Λ_k.

Imported finite-cutoff UOM analytic/structural inputs. The following finite-cutoff facts are imported from the UOM finite-cutoff package.

1. By Theorem 4.3, the renormalized Schwinger–Keldysh functional $W_{\mathrm{ren}}[J_+, J_-]$ is C^2 near the physical diagonal $J_a = 0$ for compactly supported admissible source multiplets on $\mathscr{X}_k$.
2. By Theorem 4.3, the UOM-P1–P5 package holds at finite cutoff on a sufficiently local gauge-invariant class $\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}$, including well-definedness, gauge independence, and closure of the Peierls bracket.
3. By Proposition 4.11, the receiver anomaly/Wess–Zumino monotone sector is represented by compact smearings of local scalar densities on the receiver head and hence lies in $\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}$.
4. By Definition 4.1, Proposition 4.1, and Theorem 4.1, the finite-cutoff SK state space carries a diagonal embedding of the physical sector and a swap involution $s : \mathcal{S}_{\mathrm{SK}} \to \mathcal{S}_{\mathrm{SK}}$ fixing that diagonal sector, and the equal-source locus is swap-invariant.

Assumption 1.1 (Local radial chart on the accepted branch)**.** On the accepted branch through Λ_k, choose a local continuous radial chart u near the point under consideration. All source-dependent receiver data used below depend differentiably enough on this branchwise chart that the infinitesimal geometric radial differential

$$\mathfrak{D}_u = \mathcal{L}_{c_u \partial_u}$$

is defined on the chosen slab-supported sufficiently local class.

2 Finite-Cutoff Two-Time SK Bicomplex

2.1 Local cochains on a tick slab

Definition 2.1 (Slab-supported sufficiently local cochains)**.** Let $\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k) \subset \mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}$ denote the vector space of sufficiently local gauge-invariant functionals whose support is contained in $\mathscr{X}_k$. Its monotone subspace

$$\mathcal{F}_{\mathrm{s.loc}}{}_{\mathrm{mon}}^{\Lambda_k}(\mathscr{X}_k) \subset \mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k)$$

consists of finite linear combinations of compact smearings of receiver-head scalar densities from the anomaly/Wess–Zumino monotone sector.

Definition 2.2 (Geometric odd differentials on slab-supported cochains)**.** Introduce constant Grassmann-odd parameters c_t, c_u. On source-dependent local receiver-head densities, define

$$Q_t := \mathcal{L}_{c_t \partial_t}, \qquad \mathfrak{D}_u := \mathcal{L}_{c_u \partial_u},$$

extended to $\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k)$ by the graded Leibniz rule and linearity. The action is taken on the *full smeared density*, including compact test kernels.

Lemma 2.3 (Bicomplex identities)**.** *The operators Q_t and $\mathfrak{D}_u$ are odd derivations on $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k)$ satisfying*

$$Q_t^2 = 0, \qquad \mathfrak{D}_u^2 = 0, \qquad Q_t\mathfrak{D}_u + \mathfrak{D}_u Q_t = 0.$$

Proof. The odd vector fields $c_t\partial_t$ and $c_u\partial_u$ square to zero because $c_t^2 = c_u^2 = 0$, and they commute because $[\partial_t, \partial_u] = 0$ on the local (t, u)-chart of the receiver-head data. Lie derivatives along commuting square-zero odd vector fields therefore satisfy the stated identities. □

Definition 2.4 (Physical SK sector)**.** A state is called *physical SK-admissible on $\mathscr{X}_k$* if it lies on the equal-source SK diagonal, uses the UOM renormalized variational principle, and all source insertions are compactly supported inside $\mathscr{X}_k$ so that no endpoint term arises from the closed t-contour.

Proposition 2.5 (Finite-cutoff t-Ward identity on the slab)**.** *For every physical SK-admissible state ρ and every $X \in \mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k)$,*

$$\langle Q_t X \rangle_\rho = 0.$$

Proof. The Q_t-identity is the usual closed-time-path change-of-variables argument on the equal-source SK locus of Theorem 4.1, applied to the renormalized functional supplied by Theorem 4.3. Because the time contour is closed and the source support is contained strictly inside $\mathscr{X}_k$, no endpoint term survives, so Q_t-exact insertions decouple in expectation. □

2.2 Local ambiguity quotient

Definition 2.6 (UOM local ambiguity relation on the slab)**.** On $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k)$ define an equivalence relation $\sim_{\mathrm{loc}}$ generated by:

1. Q_t-exact shifts $F \mapsto F + Q_t X$;
2. boundary-admissible total divergences $F \mapsto F + \partial_\mu K^\mu$ whose renormalized flux vanishes on the slab boundary;
3. admissible local counterterm shifts $F \mapsto F + \Delta_{\mathrm{ct}}$ whose finite-cutoff physical Peierls response vanishes on the receiver-head sector under consideration.

Write $[F]_{\mathrm{loc}}$ for the $\sim_{\mathrm{loc}}$-class of F.

Remark 2.7 (Fixed-scheme scope). The identities in propositions 3.3 and 3.5 are fixed-scheme statements on the chosen monotone sector. Inter-scheme comparison is treated separately in theorem 4.10.

3 Good Monotone Representatives on the Receiver Head

Definition 3.1 (Monotone representative by head smearing)**.** Let $L_{\mathrm{mon}}(t, x; u)$ be a receiver-head scalar density in the UOM anomaly/Wess–Zumino monotone sector. For a compactly supported spatial kernel $\eta_k \in C_c^\infty(\Sigma_{\mathrm{rec}})$ and a temporal kernel $\kappa_k \in C_c^\infty(I_k)$, define

$$\mathcal{O}_k^{(0)}(u) := \int_{\mathscr{X}_k} dt\, d^3x\, \kappa_k(t)\, \eta_k(x)\, L_{\mathrm{mon}}(t, x; u). \tag{1}$$

Proposition 3.2 (Q_t-closed head smearing)**.** *Assume $\kappa_k \in C_c^\infty(I_k)$ and that κ_k vanishes in a neighborhood of both endpoints $t_k^\pm$. Then*

$$Q_t \mathcal{O}_k^{(0)}(u) = 0.$$

Proof. Because Q_t acts on the full smeared density, including the kernel,

$$Q_t \mathcal{O}_k^{(0)}(u) = \int_{\mathscr{X}_k} dt\, d^3x\, \mathcal{L}_{c_t \partial_t}\big(\kappa_k(t)\eta_k(x) L_{\mathrm{mon}}(t,x;u)\, dt \wedge d^3x\big).$$

Since $L_{\mathrm{mon}}\, dt \wedge d^3x$ is a scalar-density integrand on the receiver head, the Lie derivative is a total t-derivative:

$$Q_t \mathcal{O}_k^{(0)}(u) = \int_{\mathscr{X}_k} \partial_t\big(c_t\, \kappa_k(t)\eta_k(x) L_{\mathrm{mon}}(t,x;u)\big)\, dt\, d^3x.$$

The compact support and endpoint vanishing of κ_k remove the slab-end contribution exactly, so the integral vanishes without passing to the local ambiguity quotient. □

Proposition 3.3 (Fixed-scheme radial Ward decomposition on the monotone sector)**.** *Let Q_u denote the full fixed-scheme radial Ward differential on $(\mathcal{F}_{\mathrm{s.loc}}{}^{\Lambda_k}_{\mathrm{mon}}(\mathscr{X}_k))$. Then, in the chosen finite-cutoff renormalization scheme,*

$$Q_u = Q_u^{\mathrm{phys}} + Q_u^{\mathrm{scheme}}, \qquad Q_u^{\mathrm{scheme}} = 0.$$

Moreover, for every Q_t-closed $X \in \mathcal{F}_{\mathrm{s.loc}}{}^{\Lambda_k}_{\mathrm{mon}}(\mathscr{X}_k)$,

$$[Q_u^{\mathrm{scheme}} X]_{\mathrm{loc}} = 0 \qquad \textit{and} \qquad [Q_u^{\mathrm{scheme}} X] = 0 \in H^1(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k), Q_t).$$

Proof. The analysis is carried out in one fixed finite-cutoff renormalization scheme, so there is no independent scheme variation inside the present note. Accordingly, the full radial Ward differential Q_u on the chosen monotone sector reduces to its physical part, and the scheme part vanishes. The two displayed triviality statements are therefore immediate because $Q_u^{\mathrm{scheme}} = 0$. □

Proposition 3.4 (Physical radial Ward/geometric identification on compact head smearings)**.** *For every compactly smeared monotone head observable in $(\mathcal{F}_{\mathrm{s.loc}}{}^{\Lambda_k}_{\mathrm{mon}}(\mathscr{X}_k))$, the physical radial Ward differential is identified with the geometric branchwise u-derivative by*

$$Q_u^{\mathrm{phys}} X = -\mathfrak{D}_u X.$$

Consequently,

$$Q_t Q_u^{\mathrm{phys}} + Q_u^{\mathrm{phys}} Q_t = 0, \qquad Q_u^{\mathrm{phys}} X = -\mathfrak{D}_u X$$

for every such observable X.

Proof. The compactly smeared scalar class used here is the slab-supported restriction of the class covered by Proposition 4.4. That result identifies the physical radial Ward differential with the geometric u-derivative on compact head smearings. In the present two-time complex, $\mathfrak{D}_u$ is the odd geometric lift of that derivative acting on the full smeared density, so the same identity reads $Q_u^{\mathrm{phys}} X = -\mathfrak{D}_u X$ on the slab-supported monotone sector. The anticommutation with Q_t follows from lemma 2.3, since for any such observable X,

$$Q_t Q_u^{\mathrm{phys}} X + Q_u^{\mathrm{phys}} Q_t X = -\left(Q_t \mathfrak{D}_u X + \mathfrak{D}_u Q_t X\right) = 0,$$

proving the claim. □

Proposition 3.5 (Imported fixed-scheme radial Ward identity on monotone head observables)**.** *For every compactly smeared monotone head observable $X(u)$,*

$$\frac{d}{du} \langle X(u) \rangle_{\rho_u} = -\langle Q_u X(u) \rangle_{\rho_u} = -\left\langle Q_u^{\mathrm{phys}} X(u) \right\rangle_{\rho_u}.$$

If $X(u)$ lies in the compactly smeared head-observable class of proposition 3.4, then equivalently

$$\frac{d}{du} \langle X(u) \rangle_{\rho_u} = \langle \mathfrak{D}_u X(u) \rangle_{\rho_u}.$$

Proof. This is the fixed-scheme radial Ward identity for the chosen branchwise renormalized expectation functional on the monotone sector. The second equality uses proposition 3.3. On compact head observables, the geometric form is the slab-supported version of Proposition 4.4 and agrees with proposition 3.4. □

Remark 3.6 (Sign convention). This note uses the convention

$$Q_u^{\mathrm{phys}} = -\mathfrak{D}_u$$

on the fixed-scheme monotone sector, so that the monotone derivative identity reads

$$\partial_u M(u) = -\left\langle Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) \right\rangle_{\rho_u}.$$

Theorem 3.7 (Good monotone representative theorem)**.** *Let $M(u)$ be a finite-cutoff UOM receiver monotone realized by (1) and suppose $M(u_0) \neq 0$ at the chosen UV-side endpoint u_0. Then there exists a sufficiently local representative $\mathcal{O}_k^{(0)}(u) \in \mathcal{F}_{\mathrm{s.loc}}{}_{\mathrm{mon}}^{\Lambda_k}(\mathscr{X}_k)$ such that:*

$$M(u) = \left\langle \mathcal{O}_k^{(0)}(u) \right\rangle_{\rho_u}, \qquad Q_t \mathcal{O}_k^{(0)}(u) = 0, \qquad [\mathcal{O}_k^{(0)}(u_0)] \neq 0 \in H^0(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k), Q_t).$$

Proof. By Proposition 4.11, the compactly smeared representative $\mathcal{O}_k^{(0)}(u)$ belongs to $(\mathcal{F}_{\mathrm{s.loc}}{}_{\mathrm{mon}}^{\Lambda_k}(\mathscr{X}_k))$. Proposition 3.2 gives $Q_t \mathcal{O}_k^{(0)}(u) = 0$.

It remains to prove nontriviality in H^0. If $\mathcal{O}_k^{(0)}(u_0) = Q_t X$ were Q_t-exact, then proposition 2.5 would give

$$M(u_0) = \left\langle \mathcal{O}_k^{(0)}(u_0) \right\rangle_{\rho_{u_0}} = \langle Q_t X \rangle_{\rho_{u_0}} = 0,$$

contradicting the assumption $M(u_0) \neq 0$. Hence $[\mathcal{O}_k^{(0)}(u_0)] \neq 0$ in degree zero cohomology. □

4 Radial Anomaly Insertion and Raw First Descent

Definition 4.1 (Radial anomaly class)**.** Let $\mathcal{O}_k^{(0)}(u)$ be a good monotone representative as in theorem 3.7. Define the odd cohomology class

$$[\mathcal{A}_k^{(1)}(u)] := [Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u)] \in H^1(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k), Q_t). \tag{2}$$

Lemma 4.2 (Monotone derivative identity in the fixed scheme)**.** *Assume assumption 1.1. For every good monotone representative $\mathcal{O}_k^{(0)}(u)$,*

$$\partial_u M(u) = -\left\langle Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) \right\rangle_{\rho_u}, \qquad \left\langle Q_u^{\mathrm{scheme}} \mathcal{O}_k^{(0)}(u) \right\rangle_{\rho_u} = 0.$$

Proof. Apply proposition 3.5 to $X(u) = \mathcal{O}_k^{(0)}(u)$. The second displayed identity then follows from proposition 3.3. □

Theorem 4.3 (Locality of differentiated radial insertion on compact head smearings)**.** *For every good monotone representative $\mathcal{O}_k^{(0)}(u)$, the differentiated radial insertion*

$$\mathcal{A}_k^{(1)}(u) := Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u)$$

defines a sufficiently local odd cochain in $(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{1}}$.

Proof. Write $\mathcal{O}_k^{(0)}(u)$ in the form (1). Because $L_{\mathrm{mon}}(t,x;u)$ is a local receiver-head scalar density and the branchwise u-chart is differentiable by assumption 1.1, the derivative $\partial_u L_{\mathrm{mon}}(t,x;u)$ is again a local scalar density depending on some finite jet order. Therefore

$$\mathcal{A}_k^{(1)}(u) = Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u) = -\int_{\mathscr{X}_k} dt\, d^3x\, \kappa_k(t)\eta_k(x)\, \partial_u L_{\mathrm{mon}}(t,x;u)$$

is a compact smearing of a local density on the receiver head, hence belongs to $(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{1}}$ by Proposition 4.11. □

Proposition 4.4 (Radial anomaly cohomology on compact head smearings)**.** *Let* $\mathcal{A}_k^{(1)}(u) := Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u) \in (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{1}}$ *be the sufficiently local odd cochain provided by theorem 4.3. Then*

$$[\mathcal{A}_k^{(1)}(u)] = [Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u)] \in H^1(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k), Q_t), \qquad Q_t\mathcal{A}_k^{(1)}(u) = 0.$$

Proof. Theorem 4.3 already gives the existence and sufficient locality of $\mathcal{A}_k^{(1)}(u)$. The cohomology identity is tautological from the definition $\mathcal{A}_k^{(1)}(u) := Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u)$. Moreover, $Q_t\mathcal{O}_k^{(0)}(u) = 0$ by proposition 3.2, and $Q_tQ_u^{\mathrm{phys}} + Q_u^{\mathrm{phys}}Q_t = 0$ by proposition 3.4, so

$$Q_t\mathcal{A}_k^{(1)}(u) = Q_tQ_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u) = -Q_u^{\mathrm{phys}}Q_t\mathcal{O}_k^{(0)}(u) = 0.$$

□

Theorem 4.5 (Canonical fixed-scheme exact anomalous descent)**.** *Assume assumption 1.1 and let* $\mathcal{O}_k^{(0)}(u)$ *be a good monotone representative. Then the canonical sufficiently local cochains are*

$$\mathcal{A}_k^{(1)}(u) := Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u) \in (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{1}}, \qquad \Xi_k^{(1)}(u) := 0 \in (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}}$$

and they satisfy

$$Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u) + Q_t\Xi_k^{(1)}(u) = \mathcal{A}_k^{(1)}(u), \qquad Q_t\mathcal{A}_k^{(1)}(u) = 0. \tag{3}$$

Proof. The first identity in (3) is immediate from the canonical choice $\mathcal{A}_k^{(1)}(u) := Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u)$, $\Xi_k^{(1)}(u) := 0$, while $Q_t\mathcal{A}_k^{(1)}(u) = 0$ is exactly proposition 4.4. □

Corollary 4.6 (Raw first descent in the local ambiguity quotient)**.** *Under the hypotheses of theorem 4.5,*

$$[Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u) + Q_t\Xi_k^{(1)}(u) - \mathcal{A}_k^{(1)}(u)]_{\mathrm{loc}} = 0.$$

Proof. This is the image of (3) in the quotient by $\sim_{\mathrm{loc}}$. □

Proposition 4.7 (Solution torsor for the exact descent equation)**.** *Fix a good monotone representative* $\mathcal{O}_k^{(0)}(u)$ *and the canonical anomaly representative* $\mathcal{A}_k^{(1)}(u) := Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u)$*. Then an even sufficiently local cochain* $\Xi_k^{(1)}(u) \in (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}}$ *solves the exact descent equation*

$$Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u) + Q_t\Xi_k^{(1)}(u) = \mathcal{A}_k^{(1)}(u)$$

if and only if

$$\Xi_k^{(1)}(u) \in \ker(Q_t) \cap (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}}.$$

Hence the solution space of the canonical exact descent equation is precisely the vector space

$$\ker(Q_t) \cap (\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathscr{X}_k))_{\bar{0}}.$$

More generally, if $\widetilde{\mathcal{A}^{(1)}}_k(u) \in (\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathscr{X}_k))_{\bar{1}}$ *is any odd representative of the same radial anomaly class as* $\mathcal{A}^{(1)}_k(u)$*, then the set of solutions of*

$$Q^{\text{phys}}_u \mathcal{O}^{(0)}_k(u) + Q_t \widetilde{\Xi^{(1)}}_k(u) = \widetilde{\mathcal{A}^{(1)}}_k(u)$$

is a nonempty affine space over

$$\ker(Q_t) \cap (\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathscr{X}_k))_{\bar{0}}.$$

Proof. For the canonical anomaly representative, the displayed equation is equivalent to

$$Q_t \Xi^{(1)}_k(u) = 0,$$

which proves the first statement.

For the general representative $\widetilde{\mathcal{A}^{(1)}}_k(u)$, class equality $[\widetilde{\mathcal{A}^{(1)}}_k(u)] = [\mathcal{A}^{(1)}_k(u)]$ implies that the defect $\widetilde{\mathcal{A}^{(1)}}_k(u) - \mathcal{A}^{(1)}_k(u)$ is Q_t-exact. Hence at least one solution $\widetilde{\Xi^{(1)}}_{k,0}(u)$ exists. If $\widetilde{\Xi^{(1)}}_k(u)$ is any other solution, subtracting the two descent equations gives

$$Q_t\big(\widetilde{\Xi^{(1)}}_k(u) - \widetilde{\Xi^{(1)}}_{k,0}(u)\big) = 0.$$

Conversely, adding any Q_t-closed even cochain to a given solution preserves the equation. Thus the full solution set is the affine space

$$\widetilde{\Xi^{(1)}}_{k,0}(u) + \ker(Q_t) \cap (\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathscr{X}_k))_{\bar{0}}.$$

□

Proposition 4.8 (Nontrivial degree-one class from nonzero RG slope)**.** *Assume the hypotheses of theorem 4.5. If*

$$\partial_u M(u)\big|_{u_0} \neq 0,$$

then

$$[Q^{\text{phys}}_u \mathcal{O}^{(0)}_k(u_0)] \neq 0 \qquad \text{in} \qquad H^1(\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathscr{X}_k), Q_t).$$

Proof. Suppose $[Q^{\text{phys}}_u \mathcal{O}^{(0)}_k(u_0)] = 0$. Then $Q^{\text{phys}}_u \mathcal{O}^{(0)}_k(u_0) = Q_t Y$ for some sufficiently local Y. Taking expectations in a physical SK-admissible state and using proposition 2.5,

$$\left\langle Q^{\text{phys}}_u \mathcal{O}^{(0)}_k(u_0) \right\rangle_{\rho_{u_0}} = \langle Q_t Y \rangle_{\rho_{u_0}} = 0.$$

By lemma 4.2, this forces $\partial_u M(u_0) = 0$, a contradiction. □

4.1 Inter-scheme comparison of the radial anomaly class

Definition 4.9 (Cohomologically admissible inter-scheme comparison)**.** Fix a good monotone representative $\mathcal{O}_k^{(0)}(u)$ on the chosen slab. Two finite-cutoff renormalization schemes S and S' at the same cutoff Λ_k are said to be *cohomologically admissibly comparable on the chosen monotone sector* if:

1. the same compactly smeared monotone observable $\mathcal{O}_k^{(0)}(u)$ belongs to $(\mathcal{F}_{\mathrm{s.loc}}{}_{\mathrm{mon}}^{\Lambda_k}(\mathscr{X}_k))$ in both schemes and remains Q_t-closed;

2. each scheme defines a sufficiently local physical radial differential on compactly smeared monotone head observables, written
$$Q_{u,S}^{\mathrm{phys}}, \qquad Q_{u,S'}^{\mathrm{phys}},$$
and the associated anomaly representatives
$$\mathcal{A}_{k,S}^{(1)}(u) := Q_{u,S}^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u), \qquad \mathcal{A}_{k,S'}^{(1)}(u) := Q_{u,S'}^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u)$$
lie in $(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{1}}$;

3. for every Q_t-closed $X \in \mathcal{F}_{\mathrm{s.loc}}{}_{\mathrm{mon}}^{\Lambda_k}(\mathscr{X}_k)$, the relative physical radial defect
$$\delta_{\mathrm{sch};S'\leftarrow S}X := \big(Q_{u,S'}^{\mathrm{phys}} - Q_{u,S}^{\mathrm{phys}}\big)X$$
admits a sufficiently local decomposition
$$\delta_{\mathrm{sch};S'\leftarrow S}X = Q_tY_X + \partial_\mu K_X^\mu + \Delta_{\mathrm{ct},X},$$
with $Y_X \in (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}}$, K_X^μ boundary-admissible on the slab, and $\Delta_{\mathrm{ct},X}$ an admissible local counterterm shift in the sense of definition 2.6; and, in addition,
$$[\delta_{\mathrm{sch};S'\leftarrow S}X] = 0 \qquad \text{in} \qquad H^1(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k), Q_t).$$

Theorem 4.10 (Inter-scheme invariance of the radial anomaly class)**.** *Assume assumption 1.1, and let S and S' be cohomologically admissibly comparable in the sense of definition 4.9. Then there exist sufficiently local data*
$$Y_k(u) \in (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}}, \qquad K_k^\mu(u), \qquad \Delta_{\mathrm{ct},k}(u)$$
such that
$$\mathcal{A}_{k,S'}^{(1)}(u) - \mathcal{A}_{k,S}^{(1)}(u) = Q_tY_k(u) + \partial_\mu K_k^\mu(u) + \Delta_{\mathrm{ct},k}(u). \tag{4}$$
Consequently,
$$[\mathcal{A}_{k,S'}^{(1)}(u)] = [\mathcal{A}_{k,S}^{(1)}(u)] \qquad \text{in} \qquad H^1(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k), Q_t),$$
and
$$[\mathcal{A}_{k,S'}^{(1)}(u) - \mathcal{A}_{k,S}^{(1)}(u)]_{\mathrm{loc}} = 0.$$
In particular, if S is the fixed scheme used in the present note, then
$$[\mathcal{A}_{k,S'}^{(1)}(u)] = [\mathcal{A}_k^{(1)}(u)] \qquad \text{in} \qquad H^1(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k), Q_t),$$
and
$$[\mathcal{A}_{k,S'}^{(1)}(u) - \mathcal{A}_k^{(1)}(u)]_{\mathrm{loc}} = 0.$$

Proof. Apply item (3) of definition 4.9 to the chosen good monotone representative $X = \mathcal{O}_k^{(0)}(u)$. Since

$$\mathcal{A}_{k,S}^{(1)}(u) = Q_{u,S}^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u), \qquad \mathcal{A}_{k,S'}^{(1)}(u) = Q_{u,S'}^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u),$$

the relative defect identity becomes exactly (4), with $Y_k(u) := Y_{\mathcal{O}_k^{(0)}(u)}$, $K_k^{\mu}(u) := K^{\mu}_{\mathcal{O}_k^{(0)}(u)}$, and $\Delta_{\mathrm{ct},k}(u) := \Delta_{\mathrm{ct},\mathcal{O}_k^{(0)}(u)}$.

The local-quotient statement follows immediately from definition 2.6, because the right-hand side of (4) is a sum of generating ambiguity moves. The cohomology statement is the specialization of the final clause of item (3) in definition 4.9 to $X = \mathcal{O}_k^{(0)}(u)$. If S is the present fixed scheme, then $\mathcal{A}_{k,S}^{(1)}(u) = \mathcal{A}_k^{(1)}(u)$ by theorem 4.5, giving the last two displays. □

Corollary 4.11 (Affine stability of descendant spaces for exactly compared representatives)**.** *Under the hypotheses of theorem 4.10, choose sufficiently local odd representatives*

$$\widehat{\mathcal{A}}_{k,S}^{(1)}(u), \qquad \widehat{\mathcal{A}}_{k,S'}^{(1)}(u)$$

of the common degree-one class $[\mathcal{A}_{k,S}^{(1)}(u)] = [\mathcal{A}_{k,S'}^{(1)}(u)]$ *such that*

$$\widehat{\mathcal{A}}_{k,S'}^{(1)}(u) - \widehat{\mathcal{A}}_{k,S}^{(1)}(u) = \mathcal{A}_{k,S'}^{(1)}(u) - \mathcal{A}_{k,S}^{(1)}(u) + Q_t Y_k(u) \tag{5}$$

for some even sufficiently local cochain $Y_k(u)$*. Let*

$$\mathrm{Sol}_S\big(\widehat{\mathcal{A}}_{k,S}^{(1)}(u)\big)$$

and

$$\mathrm{Sol}_{S'}\big(\widehat{\mathcal{A}}_{k,S'}^{(1)}(u)\big)$$

denote the affine spaces of sufficiently local exact descendants solving

$$Q_{u,S}^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) + Q_t \Xi_k^{(1)}(u) = \widehat{\mathcal{A}}_{k,S}^{(1)}(u)$$

and

$$Q_{u,S'}^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) + Q_t \Xi_k^{(1)}(u) = \widehat{\mathcal{A}}_{k,S'}^{(1)}(u),$$

respectively. Then

$$\mathrm{Sol}_{S'}\big(\widehat{\mathcal{A}}_{k,S'}^{(1)}(u)\big) = \mathrm{Sol}_S\big(\widehat{\mathcal{A}}_{k,S}^{(1)}(u)\big) + Y_k(u).$$

In particular, once the anomaly representatives are chosen in exactly compared form, the two descendant affine spaces differ only by translation by $Y_k(u)$*. If* $Y_k'(u)$ *is another primitive satisfying* (5) *for the same pair of representatives, then*

$$Y_k'(u) - Y_k(u) \in \ker(Q_t) \cap (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}}.$$

Proof. Let $\Xi_{k,S}^{(1)}(u) \in \mathrm{Sol}_S(\widehat{\mathcal{A}}_{k,S}^{(1)}(u))$ and define

$$\Xi_{k,S'}^{(1)}(u) := \Xi_{k,S}^{(1)}(u) + Y_k(u).$$

Using the exact descendant equation in scheme S and the exact comparison (5), we compute

$$Q_{u,S'}^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) + Q_t \Xi_{k,S'}^{(1)}(u) = \widehat{\mathcal{A}}_{k,S'}^{(1)}(u).$$

Thus translation by $+Y_k(u)$ sends solutions in scheme S to solutions in scheme S', and the inverse translation by $-Y_k(u)$ gives the converse inclusion. If $Y_k'(u)$ is another primitive for the same chosen pair of representatives, then

$$Q_t\big(Y_k'(u) - Y_k(u)\big) = 0,$$

which is the final claim. □

5 Conditional Hodge/Pseudoinverse Normalization

Theorem 5.1 (First-variation Hilbertization on the monotone sector)**.** *Assume the finite-cutoff UOM input package of section 1 and fix u in the branchwise neighborhood of assumption 1.1. Let $\varphi_u \in \mathcal{X}^s_{\Lambda_k}$ denote the corresponding finite-cutoff receiver-head boundary data. For each parity $\bar{\imath} \in \{\bar{0}, \bar{1}\}$, define*

$$\mathcal{Y}^{\Lambda_k}_{\mathrm{pre},\bar{\imath}}(\mathscr{X}_k; u) := \mathrm{span}\Big\{ F^{(1)}(\varphi_u) \;\Big|\; F \in (\mathcal{F}_{\mathrm{s.loc}}{}^{\Lambda_k}_{\mathrm{mon}}(\mathscr{X}_k))_{\bar{\imath}} \Big\}.$$

Then:

1. *every element of $\mathcal{Y}^{\Lambda_k}_{\mathrm{pre},\bar{\imath}}(\mathscr{X}_k; u)$ is a compactly supported band-limited receiver-head density on $\mathscr{X}_k$;*

2. *after choosing any positive-definite inner product m_{AB} on the finite component space of receiver boundary fields, the pairing*

$$(\alpha, \beta)_{\mathrm{fv},k,u} := \int_{\mathscr{X}_k} dt\, d^3x\, \sqrt{-\gamma}\, m^{AB} \alpha_A(x) \beta_B(x)$$

   defines a pre-Hilbert structure on each $\mathcal{Y}^{\Lambda_k}_{\mathrm{pre},\bar{\imath}}(\mathscr{X}_k; u)$;

3. *if $\mathcal{Y}^{\Lambda_k}_{\bar{\imath}}(\mathscr{X}_k; u)$ denotes the corresponding Hilbert completion and*

$$\mathcal{N}_{\bar{\imath},k}(u) := \big\{ F \in (\mathcal{F}_{\mathrm{s.loc}}{}^{\Lambda_k}_{\mathrm{mon}}(\mathscr{X}_k))_{\bar{\imath}} \;:\; F^{(1)}(\varphi_u) = 0 \big\},$$

   then the first-variation map descends to an injective linear map

$$\jmath_{\bar{\imath},u} : (\mathcal{F}_{\mathrm{s.loc}}{}^{\Lambda_k}_{\mathrm{mon}}(\mathscr{X}_k))_{\bar{\imath}} / \mathcal{N}_{\bar{\imath},k}(u) \hookrightarrow \mathcal{Y}^{\Lambda_k}_{\bar{\imath}}(\mathscr{X}_k; u)$$

   with dense image.

*In particular, the selected monotone sector admits a theorem-level first-variation Hilbertization, which is the natural analytic input underlying the Hodge target **(H1)**.*

Proof. By Definition 4.7 and (148), every $F \in \mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k)$ is a finite sum of compactly supported smearings of local jet polynomials in the band-limited receiver-head variables, and its first Fréchet derivative $F^{(1)}(\varphi_u)$ exists as a compactly supported density-valued function on the receiver head. For the monotone sector this derivative is again supported in $\mathscr{X}_k$, and band limitation is preserved because the receiver data lie in the projected space $\mathcal{X}^s_{\Lambda_k}$. Finite linear combinations therefore remain compactly supported band-limited densities, proving item (1).

Because the receiver cutoff leaves only finitely many field components and because the slab is bounded, a choice of positive-definite fiber metric m_{AB} produces a positive-definite L^2-type pairing on such density coefficients. Compact support of α, β makes the integral finite, and bilinearity is immediate, proving item (2).

For item (3), quotient by the null space of the first-variation map so that the induced map $\jmath_{\bar{\imath},u}$ is injective. Its image contains $\mathcal{Y}^{\Lambda_k}_{\mathrm{pre},\bar{\imath}}(\mathscr{X}_k; u)$ by construction, hence is dense in the Hilbert completion $\mathcal{Y}^{\Lambda_k}_{\bar{\imath}}(\mathscr{X}_k; u)$. □

Definition 5.2 (Admissible endpoint-vanishing time-kernel sector)**.** Fix u and choose a real Hilbert space $\mathcal{R}_k(u)$ representing the finite-cutoff band-limited receiver coefficient sector on Σ_{rec}. Define the Bochner spaces

$$\mathcal{K}^{\Lambda_k}_{\bar{0},k}(u) := H^1_0(I_k;\mathcal{R}_k(u)), \qquad \mathcal{K}^{\Lambda_k}_{\bar{1},k}(u) := L^2(I_k;\mathcal{R}_k(u)),$$

and the zero-mode-free odd sector

$$\mathcal{K}^{\Lambda_k,0}_{\bar{1},k}(u) := \left\{ G \in L^2(I_k;\mathcal{R}_k(u)) \;\middle|\; \int_{I_k} G(t)\,dt = 0 \right\},$$

where the integral is the Bochner integral in $\mathcal{R}_k(u)$. The kernel-level t-differential is

$$\overline{Q_t}_{\mathrm{ker},k} := c_t\partial_t : \mathcal{K}^{\Lambda_k}_{\bar{0},k}(u) \to \mathcal{K}^{\Lambda_k}_{\bar{1},k}(u),$$

with the odd parameter c_t recording parity and ∂_t the underlying real operator.

Theorem 5.3 (Closedness and closed range of Q_t on the admissible time-kernel sector)**.** *In the setting of definition 5.2, the operator*

$$\partial_t : H^1_0(I_k;\mathcal{R}_k(u)) \to L^2(I_k;\mathcal{R}_k(u))$$

is densely defined and closed, and its range is exactly

$$\mathrm{ran}(\partial_t) = \mathcal{K}^{\Lambda_k,0}_{\bar{1},k}(u).$$

Hence $\mathrm{ran}(\partial_t)$ *is a closed subspace of* $L^2(I_k;\mathcal{R}_k(u))$*. Equivalently,*

$$\mathrm{ran}(\overline{Q_t}_{\mathrm{ker},k}) = c_t\,\mathcal{K}^{\Lambda_k,0}_{\bar{1},k}(u)$$

is closed in the odd copy of the admissible time-kernel sector. Moreover, there is a Poincaré–Friedrichs estimate

$$\|F\|_{L^2(I_k;\mathcal{R}_k(u))} \le |I_k|\,\|\partial_t F\|_{L^2(I_k;\mathcal{R}_k(u))}$$

for all $F \in H^1_0(I_k;\mathcal{R}_k(u))$*.*

Proof. Density of the domain is standard because $C^\infty_c(I_k;\mathcal{R}_k(u))$ is dense in $H^1_0(I_k;\mathcal{R}_k(u))$, and hence dense in $L^2(I_k;\mathcal{R}_k(u))$.

To prove closedness, let $F_n \in H^1_0(I_k;\mathcal{R}_k(u))$ satisfy

$$F_n \to F \quad \text{in} \quad L^2(I_k;\mathcal{R}_k(u)), \qquad \partial_t F_n \to G \quad \text{in} \quad L^2(I_k;\mathcal{R}_k(u)).$$

For every scalar test function $\phi \in C^\infty_c(I_k)$ and every $v \in \mathcal{R}_k(u)$, integration by parts gives

$$\int_{I_k} \langle F_n(t), v\rangle\,\phi'(t)\,dt = -\int_{I_k} \langle \partial_t F_n(t), v\rangle\,\phi(t)\,dt.$$

Passing to the limit shows that F has weak derivative G, hence $F \in H^1(I_k;\mathcal{R}_k(u))$ and $\partial_t F = G$. Because the trace map $H^1(I_k;\mathcal{R}_k(u)) \to \mathcal{R}_k(u) \times \mathcal{R}_k(u)$ is continuous and each F_n has vanishing endpoint traces, F also has vanishing endpoint traces. Therefore $F \in H^1_0(I_k;\mathcal{R}_k(u))$, proving that ∂_t is closed.

If $F \in H^1_0(I_k;\mathcal{R}_k(u))$, the Bochner fundamental theorem of calculus yields

$$\int_{I_k} \partial_t F(t)\,dt = F(t_k^+) - F(t_k^-) = 0,$$

so $\operatorname{ran}(\partial_t) \subset \mathcal{K}^{\Lambda_k,0}_{\bar{1},k}(u)$. Conversely, let $G \in \mathcal{K}^{\Lambda_k,0}_{\bar{1},k}(u)$ and define

$$(\mathsf{S}_k G)(t) := \int_{t_k^-}^{t} G(s)\, ds.$$

Then $\mathsf{S}_k G \in H^1(I_k; \mathcal{R}_k(u))$, $\partial_t(\mathsf{S}_k G) = G$ a.e., and

$$(\mathsf{S}_k G)(t_k^-) = 0, \qquad (\mathsf{S}_k G)(t_k^+) = \int_{I_k} G(s)\, ds = 0,$$

so $\mathsf{S}_k G \in H^1_0(I_k; \mathcal{R}_k(u))$. Hence $\operatorname{ran}(\partial_t) = \mathcal{K}^{\Lambda_k,0}_{\bar{1},k}(u)$, and this range is closed because the zero-mode-free Bochner subspace is closed in L^2.

Finally, for $F \in H^1_0(I_k; \mathcal{R}_k(u))$,

$$F(t) = \int_{t_k^-}^{t} \partial_t F(s)\, ds,$$

so

$$\|F(t)\|_{\mathcal{R}_k(u)} \le \int_{I_k} \|\partial_t F(s)\|_{\mathcal{R}_k(u)}\, ds.$$

Integrating in t and applying Cauchy–Schwarz gives

$$\|F\|_{L^2(I_k;\mathcal{R}_k(u))} \le |I_k|\, \|\partial_t F\|_{L^2(I_k;\mathcal{R}_k(u))},$$

which is the stated Poincaré–Friedrichs estimate. □

Assumption 5.4 (Conditional analytic package beyond theorem-level H1–H2)**.** Fix the first-variation Hilbertizations

$$\mathcal{Y}^{\Lambda_k}_{\bar{0}}(\mathscr{X}_k; u), \qquad \mathcal{Y}^{\Lambda_k}_{\bar{1}}(\mathscr{X}_k; u)$$

of theorem 5.1. Assume in addition that there exist real Hilbert spaces

$$\mathcal{H}^{\Lambda_k}_{\bar{0}}(\mathscr{X}_k), \qquad \mathcal{H}^{\Lambda_k}_{\bar{1}}(\mathscr{X}_k),$$

giving a compatible raw-cochain realization of that first-variation Hilbertization, with continuous dense inclusions

$$(\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathscr{X}_k))_{\bar{0}} \hookrightarrow \mathcal{H}^{\Lambda_k}_{\bar{0}}(\mathscr{X}_k), \qquad (\mathcal{F}^{\Lambda_k}_{\text{s.loc}}(\mathscr{X}_k))_{\bar{1}} \hookrightarrow \mathcal{H}^{\Lambda_k}_{\bar{1}}(\mathscr{X}_k),$$

such that:

1. **Compatibility with the theorem-level time-kernel sector.** There are closed subspaces

   $$\mathcal{H}^{\Lambda_k}_{\bar{0},\text{adm}}(\mathscr{X}_k; u) \subset \mathcal{H}^{\Lambda_k}_{\bar{0}}(\mathscr{X}_k), \qquad \mathcal{H}^{\Lambda_k}_{\bar{1},\text{adm}}(\mathscr{X}_k; u) \subset \mathcal{H}^{\Lambda_k}_{\bar{1}}(\mathscr{X}_k),$$

   unitarily identified with the admissible time-kernel sector of definition 5.2 as the closures of the corresponding raw sufficiently local kernel-generated sectors. Writing

   $$\overline{Q_t}_{\text{adm}} : \mathcal{D}(\overline{Q_t}_{\text{adm}}) \subset \mathcal{H}^{\Lambda_k}_{\bar{0},\text{adm}}(\mathscr{X}_k; u) \to \mathcal{H}^{\Lambda_k}_{\bar{1},\text{adm}}(\mathscr{X}_k; u)$$

   for the restriction of $\overline{Q_t}$ to those subspaces, one has

   $$\overline{Q_t}_{\text{adm}} \cong \overline{Q_t}_{\text{ker},k}$$

   under the chosen unitary identifications.

2. **Branch-symmetric norm choice.** The dense sufficiently local sectors are Γ-stable, and the chosen Hilbert norms satisfy

$$\|\Gamma F\|_{\mathcal{H}_{\bar{0}}^{\Lambda_k}(\mathscr{X}_k)} = \|F\|_{\mathcal{H}_{\bar{0}}^{\Lambda_k}(\mathscr{X}_k)}, \qquad \|\Gamma G\|_{\mathcal{H}_{\bar{1}}^{\Lambda_k}(\mathscr{X}_k)} = \|G\|_{\mathcal{H}_{\bar{1}}^{\Lambda_k}(\mathscr{X}_k)}$$

for all $F \in (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}}$ and $G \in (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{1}}$.

3. **(H4) Admissible representative class and return problem.** There is a selected class

$$\mathcal{C}_{\mathrm{adm},k}(u) \subset (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{1}}$$

of sufficiently local odd representatives of the radial anomaly class $[\mathcal{A}_k^{(1)}(u)]$ such that every $\widetilde{\mathcal{A}^{(1)}}_k(u) \in \mathcal{C}_{\mathrm{adm},k}(u)$ lies in $\mathcal{H}_{\bar{1},\mathrm{adm}}^{\Lambda_k}(\mathscr{X}_k; u)$, its defect

$$\widetilde{B}_k(u) := \widetilde{\mathcal{A}^{(1)}}_k(u) - \mathcal{A}_k^{(1)}(u)$$

belongs to $\mathrm{ran}(\overline{Q_t}_{\mathrm{adm}})$. The raw return problem is then to show, on this selected class, that the Moore–Penrose-selected primitive of $\widetilde{B}_k(u)$ lies back in $(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}}$.

Remark 5.5 (Scope of the Hodge section). The raw package of theorem 6.9 is independent of the present section. The additional hypotheses in assumption 5.4 are used only to choose and normalize representatives of the same degree-one class on the selected admissible Hilbert sector.

Theorem 5.6 (Branch-symmetric Hilbert extension of Γ)**.** *Assume assumption 5.4. Then:*

1. *the dense raw action of Γ extends uniquely to bounded isometric involutions on $\mathcal{H}_{\bar{0}}^{\Lambda_k}(\mathscr{X}_k)$ and $\mathcal{H}_{\bar{1}}^{\Lambda_k}(\mathscr{X}_k)$;*

2. *the admissible closed subspaces $\mathcal{H}_{\bar{0},\mathrm{adm}}^{\Lambda_k}(\mathscr{X}_k; u)$ and $\mathcal{H}_{\bar{1},\mathrm{adm}}^{\Lambda_k}(\mathscr{X}_k; u)$ are Γ-stable;*

3. *on the admissible closed sector, the domain of $\overline{Q_t}_{\mathrm{adm}}$ is Γ-stable and*

$$\Gamma \overline{Q_t}_{\mathrm{adm}} = \overline{Q_t}_{\mathrm{adm}} \Gamma;$$

4. *the domain of $\overline{Q_t}_{\mathrm{adm}}^{\dagger}$ is Γ-stable and*

$$\Gamma \overline{Q_t}_{\mathrm{adm}}^{\dagger} = \overline{Q_t}_{\mathrm{adm}}^{\dagger} \Gamma;$$

5. *writing*

$$\overline{Q_t}_{\mathrm{adm}}^{+} := \overline{Q_t}_{\mathrm{adm}}^{\dagger} \big(\overline{Q_t}_{\mathrm{adm}}\, \overline{Q_t}_{\mathrm{adm}}^{\dagger}\big)^{-1}$$

for the Moore–Penrose inverse on the admissible closed-range sector of theorem 5.3, one has

$$\Gamma \overline{Q_t}_{\mathrm{adm}}^{+} = \overline{Q_t}_{\mathrm{adm}}^{+} \Gamma.$$

Proof. Because the Hilbert norms are Γ-invariant on the dense raw sectors, Γ is an isometry there. Every isometry on a dense subspace extends uniquely by continuity to a bounded isometric operator on the Hilbert completion, and $\Gamma^2 = \mathrm{id}$ on the dense raw sector implies $\Gamma^2 = \mathrm{id}$ after extension. This proves item (1).

Because the raw kernel-generated sectors are Γ-stable and the admissible closed subspaces are their closures, item (2) follows immediately from continuity of the Hilbert extension.

For item (3), let F lie in the domain of $\overline{Q_t}_{\mathrm{adm}}$. By construction of the admissible sector as the closure of the raw kernel-generated sector, there exists a sequence F_n in the corresponding raw sector such that

$$F_n \to F, \qquad \overline{Q_t}_{\mathrm{adm}} F_n \to \overline{Q_t}_{\mathrm{adm}} F$$

in the graph norm of $\overline{Q_t}_{\mathrm{adm}}$. On the raw sector, $\Gamma Q_t = Q_t \Gamma$ by lemma 6.2. Applying the bounded extension of Γ and passing to the limit gives

$$\Gamma F_n \to \Gamma F, \qquad \overline{Q_t}_{\mathrm{adm}} \Gamma F_n = \Gamma \overline{Q_t}_{\mathrm{adm}} F_n \to \Gamma \overline{Q_t}_{\mathrm{adm}} F.$$

Since $\overline{Q_t}_{\mathrm{adm}}$ is closed on that sector by theorem 5.3, it follows that $\Gamma F \in \mathcal{D}(\overline{Q_t}_{\mathrm{adm}})$ and $\overline{Q_t}_{\mathrm{adm}} \Gamma F = \Gamma \overline{Q_t}_{\mathrm{adm}} F$, proving item (3).

For item (4), let $G \in \mathcal{D}(\overline{Q_t}^{\dagger}_{\mathrm{adm}})$ and $F \in \mathcal{D}(\overline{Q_t}_{\mathrm{adm}})$ on the admissible closed sector. Using item (3) and the fact that Γ is an isometric involution, we compute

$$\langle \overline{Q_t}_{\mathrm{adm}} F, \Gamma G \rangle = \langle \Gamma \overline{Q_t}_{\mathrm{adm}} F, G \rangle = \langle \overline{Q_t}_{\mathrm{adm}} \Gamma F, G \rangle = \langle \Gamma F, \overline{Q_t}^{\dagger}_{\mathrm{adm}} G \rangle = \langle F, \Gamma \overline{Q_t}^{\dagger}_{\mathrm{adm}} G \rangle.$$

Therefore $\Gamma G \in \mathcal{D}(\overline{Q_t}^{\dagger}_{\mathrm{adm}})$ and $\overline{Q_t}^{\dagger}_{\mathrm{adm}} \Gamma G = \Gamma \overline{Q_t}^{\dagger}_{\mathrm{adm}} G$, proving item (4).

Finally, item (4) implies that Γ commutes with $\overline{Q_t}_{\mathrm{adm}}\, \overline{Q_t}^{\dagger}_{\mathrm{adm}}$ on its natural domain, hence also with its inverse on $\mathrm{ran}(\overline{Q_t}_{\mathrm{adm}})$ in the admissible closed-range sector of theorem 5.3. Combining this with item (4) gives

$$\begin{aligned}
\Gamma \overline{Q_t}^{+}_{\mathrm{adm}} &= \Gamma \overline{Q_t}^{\dagger}_{\mathrm{adm}} \big(\overline{Q_t}_{\mathrm{adm}}\, \overline{Q_t}^{\dagger}_{\mathrm{adm}}\big)^{-1} \\
&= \overline{Q_t}^{\dagger}_{\mathrm{adm}} \Gamma \big(\overline{Q_t}_{\mathrm{adm}}\, \overline{Q_t}^{\dagger}_{\mathrm{adm}}\big)^{-1} \\
&= \overline{Q_t}^{\dagger}_{\mathrm{adm}} \big(\overline{Q_t}_{\mathrm{adm}}\, \overline{Q_t}^{\dagger}_{\mathrm{adm}}\big)^{-1} \Gamma \\
&= \overline{Q_t}^{+}_{\mathrm{adm}} \Gamma,
\end{aligned}$$

which is item (5). □

Theorem 5.7 (Conditional Hodge/pseudoinverse normalization)**.** *Assume assumption 5.4. Let*

$$\widetilde{\mathcal{A}^{(1)}}_k(u) \in \mathcal{C}_{\mathrm{adm},k}(u)$$

be an admissible sufficiently local odd representative of the same radial anomaly class as $\mathcal{A}^{(1)}_k(u) = Q^{\mathrm{phys}}_u \mathcal{O}^{(0)}_k(u)$*, and define the representative defect*

$$\widetilde{B}_k(u) := \widetilde{\mathcal{A}^{(1)}}_k(u) - \mathcal{A}^{(1)}_k(u).$$

Then $\widetilde{B}_k(u) \in \mathrm{ran}(\overline{Q_t}_{\mathrm{adm}})$*, and the Moore–Penrose normalized primitive*

$$\widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u) := \overline{Q_t}^{\dagger}_{\mathrm{adm}} \big(\overline{Q_t}_{\mathrm{adm}}\, \overline{Q_t}^{\dagger}_{\mathrm{adm}}\big)^{-1} \widetilde{B}_k(u)$$

is the unique element of $\ker(\overline{Q_t}_{\mathrm{adm}})^{\perp} \subset \mathcal{H}^{\Lambda_k}_{\bar{0},\mathrm{adm}}(\mathscr{X}_k; u)$ *of minimal Hilbert norm satisfying*

$$\overline{Q_t}_{\mathrm{adm}}\, \widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u) = \widetilde{B}_k(u).$$

Consequently,

$$Q^{\mathrm{phys}}_u \mathcal{O}^{(0)}_k(u) + \overline{Q_t}_{\mathrm{adm}}\, \widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u) = \widetilde{\mathcal{A}^{(1)}}_k(u)$$

holds in the admissible closed Hilbert sector. If, in addition, $\widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u) \in (\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k))_{\bar{0}}$*, then the same equation holds in the raw sufficiently local complex.*

Proof. Because $\widetilde{\mathcal{A}^{(1)}}_k(u)$ and $\mathcal{A}_k^{(1)}(u)$ represent the same cohomology class, the defect $\widetilde{B}_k(u)$ is Q_t-exact in the raw complex. By the compatibility clause and **(H4)** of assumption 5.4, admissible representative defects lie in the admissible closed subspace on which $\overline{Q_t}_{\mathrm{adm}}$ agrees with the closed-range operator of theorem 5.3. Applying closed-range Hilbert-space Hodge theory to that restricted sector, and writing the restricted operator explicitly as $\overline{Q_t}_{\mathrm{adm}}$, we obtain: standard closed-range Hilbert-space Hodge theory gives the orthogonal decomposition

$$\mathcal{H}^{\Lambda_k}_{\bar{0},\mathrm{adm}}(\mathscr{X}_k;u) = \ker(\overline{Q_t}_{\mathrm{adm}}) \oplus \overline{\mathrm{ran}(\overline{Q_t}^{\dagger}_{\mathrm{adm}})},$$

and the Moore–Penrose right inverse of $\overline{Q_t}_{\mathrm{adm}}$ on $\mathrm{ran}(\overline{Q_t}_{\mathrm{adm}})$ is

$$\overline{Q_t}^{+}_{\mathrm{adm}} = \overline{Q_t}^{\dagger}_{\mathrm{adm}}\big(\overline{Q_t}_{\mathrm{adm}}\,\overline{Q_t}^{\dagger}_{\mathrm{adm}}\big)^{-1}.$$

Setting $\widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u) := \overline{Q_t}^{+}_{\mathrm{adm}}\widetilde{B}_k(u)$ therefore yields the unique solution in $\ker(\overline{Q_t}_{\mathrm{adm}})^{\perp}$ of minimal Hilbert norm. The displayed descent equation in the admissible sector is then immediate from the definition of $\widetilde{B}_k(u)$. If the normalized primitive happens to lie back in the sufficiently local complex, the same identity holds there without change. □

Corollary 5.8 (Parity of the Hodge-normalized primitive)**.** *Assume assumption 5.4. If the representative defect $\widetilde{B}_k(u)$ is Γ-odd, respectively Γ-even, then the Hodge-normalized primitive $\widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u)$ is also Γ-odd, respectively Γ-even.*

Proof. By theorem 5.6, Γ is a Hilbert isometry and commutes with $\overline{Q_t}_{\mathrm{adm}}$, $\overline{Q_t}^{\dagger}_{\mathrm{adm}}$, and the Moore–Penrose right inverse

$$\overline{Q_t}^{+}_{\mathrm{adm}} = \overline{Q_t}^{\dagger}_{\mathrm{adm}}\big(\overline{Q_t}_{\mathrm{adm}}\,\overline{Q_t}^{\dagger}_{\mathrm{adm}}\big)^{-1}$$

on $\mathrm{ran}(\overline{Q_t}_{\mathrm{adm}})$. Hence

$$\Gamma\widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u) = \Gamma\overline{Q_t}^{+}_{\mathrm{adm}}\widetilde{B}_k(u) = \overline{Q_t}^{+}_{\mathrm{adm}}\Gamma\widetilde{B}_k(u).$$

If $\Gamma\widetilde{B}_k(u) = \pm\widetilde{B}_k(u)$, the same sign is inherited by $\widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u)$. □

Corollary 5.9 (Hilbertized affine solution space)**.** *Under the hypotheses of theorem 5.7, every solution of*

$$Q_u^{\mathrm{phys}}\mathcal{O}_k^{(0)}(u) + \overline{Q_t}_{\mathrm{adm}}\,\widetilde{\Xi^{(1)}}_k(u) = \widetilde{\mathcal{A}^{(1)}}_k(u)$$

in the admissible closed Hilbert sector is of the form

$$\widetilde{\Xi^{(1)}}_k(u) = \widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u) + K_k(u), \qquad K_k(u) \in \ker(\overline{Q_t}_{\mathrm{adm}}).$$

Proof. Subtract any two solutions and apply $\overline{Q_t}_{\mathrm{adm}}$. Their difference lies in $\ker(\overline{Q_t}_{\mathrm{adm}})$, while adding a kernel element to a given solution preserves the equation. □

Remark 5.10 (Canonical representative in the Hodge sector). For the canonical representative $\widetilde{\mathcal{A}^{(1)}}_k(u) = \mathcal{A}_k^{(1)}(u)$, the defect

$$\widetilde{B}_k(u) = \widetilde{\mathcal{A}^{(1)}}_k(u) - \mathcal{A}_k^{(1)}(u)$$

vanishes. Hence the Hodge-normalized primitive furnished by theorem 5.7 is zero.

6 Swap Symmetry and Summary Theorems

Definition 6.1 (Swap action on sufficiently local cochains)**.** Let $s : \mathcal{S}_{\mathrm{SK}} \to \mathcal{S}_{\mathrm{SK}}$ be the finite-cutoff UOM swap involution of Definition 4.1 and Theorem 4.1. Define its pullback Γ on sufficiently local cochains by

$$(\Gamma F)(x) := F(sx).$$

Lemma 6.2 (Swap commutes with the two-time bicomplex)**.** *On* $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k)$*,*

$$\Gamma Q_t = Q_t \Gamma, \qquad \Gamma \mathfrak{D}_u = \mathfrak{D}_u \Gamma.$$

Proof. The swap acts only on the SK branch labels, while Q_t and $\mathfrak{D}_u$ are Lie derivatives along the base directions t and u with fixed odd parameters c_t, c_u. Therefore swap and the two odd reparametrization operators commute. □

Proposition 6.3 (Swap covariance of the fixed-scheme radial split)**.** *On the monotone sector,*

$$\Gamma Q_u^{\mathrm{phys}} = Q_u^{\mathrm{phys}} \Gamma, \qquad \Gamma Q_u^{\mathrm{scheme}} = Q_u^{\mathrm{scheme}} \Gamma.$$

Proof. By proposition 3.4, $Q_u^{\mathrm{phys}} = -\mathfrak{D}_u$ and $Q_u^{\mathrm{scheme}} = 0$ on the chosen monotone sector. The first identity is therefore

$$\Gamma Q_u^{\mathrm{phys}} = -\,\Gamma \mathfrak{D}_u = -\,\mathfrak{D}_u \Gamma = Q_u^{\mathrm{phys}} \Gamma,$$

using lemma 6.2, and the second is immediate because $Q_u^{\mathrm{scheme}} = 0$. □

Proposition 6.4 (Swap-even core cochains)**.** *Let* $\mathcal{O}_k^{(0)}(u)$ *be the compactly smeared monotone representative* (1) *on the equal-source physical sector, and let* $\mathcal{A}_k^{(1)}(u) := Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u)$*. Then*

$$\Gamma \mathcal{O}_k^{(0)}(u) = \mathcal{O}_k^{(0)}(u), \qquad \Gamma \mathcal{A}_k^{(1)}(u) = \mathcal{A}_k^{(1)}(u).$$

Proof. The kernels κ_k, η_k are ordinary branch-independent test functions, while $L_{\mathrm{mon}}(t, x; u)$ is a receiver-head scalar density evaluated on the equal-source physical SK diagonal. By Definition 4.1 and Theorem 4.1, the swap involution acts only on the SK branch labels and fixes that diagonal sector. Hence the full compact smearing defining $\mathcal{O}_k^{(0)}(u)$ is swap-invariant: $\Gamma \mathcal{O}_k^{(0)}(u) = \mathcal{O}_k^{(0)}(u)$. Using proposition 6.3, we then obtain

$$\Gamma \mathcal{A}_k^{(1)}(u) = \Gamma Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) = Q_u^{\mathrm{phys}} \Gamma \mathcal{O}_k^{(0)}(u) = Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) = \mathcal{A}_k^{(1)}(u).$$

□

Proposition 6.5 (Swap preserves anomalous descent)**.** *Assume theorem 4.5. If* $\Xi_k^{(1)}(u)$ *satisfies the exact anomalous descent equation* (3)*, then* $\Gamma \Xi_k^{(1)}(u)$ *satisfies the same exact anomalous descent equation.*

Proof. Apply Γ to (3). Using lemma 6.2 and propositions 6.3 and 6.4, we get

$$\Gamma(Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) + Q_t \Xi_k^{(1)}(u)) = Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) + Q_t \Gamma \Xi_k^{(1)}(u) = \Gamma \mathcal{A}_k^{(1)}(u) = \mathcal{A}_k^{(1)}(u).$$

This is exactly the anomalous descent equation for $\Gamma \Xi_k^{(1)}(u)$. □

Corollary 6.6 (Swap-odd descendant choice)**.** *Under the hypotheses of proposition 6.5, the odd projection*

$$\Xi^{(1)}_{-,k}(u) := \frac{1}{2}\big(\Xi^{(1)}_k(u) - \Gamma\Xi^{(1)}_k(u)\big)$$

is Γ-odd and satisfies the same exact anomalous descent equation.

Proof. Both $\Xi^{(1)}_k(u)$ and $\Gamma\Xi^{(1)}_k(u)$ solve the same exact descent equation. Subtracting and dividing by 2 preserves that equation, and the parity identity $\Gamma\Xi^{(1)}_{-,k}(u) = -\Xi^{(1)}_{-,k}(u)$ is immediate. □

Corollary 6.7 (Odd solution torsor for the canonical anomaly)**.** *For the canonical anomaly representative $\mathcal{A}^{(1)}_k(u) = Q^{\mathrm{phys}}_u \mathcal{O}^{(0)}_k(u)$, the Γ-odd exact descendants are precisely*

$$\ker(Q_t) \cap (\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k))_{\bar{0}} \cap \ker(\Gamma + \mathrm{id}).$$

Proof. By proposition 4.7, the exact descendants for the canonical anomaly equation are exactly the Q_t-closed even sufficiently local cochains. Among them, the Γ-odd ones are precisely those annihilated by $\Gamma + \mathrm{id}$. Conversely, any cochain in the displayed intersection is Q_t-closed and therefore solves the canonical exact descent equation. □

Remark 6.8 (Odd extension versus odd nonvanishing)*.* Theorem 6.11 produces swap-odd exact descendants but does not prove that any such descendant is nonzero. Nonvanishing requires an additional witness, for example the overlap and witness-comparison criteria of theorems 4.21, 4.38 and 4.45. For the canonical fixed-scheme raw solution $\Xi^{(1)}_k(u) = 0$, the odd projection vanishes identically, whereas corollary 6.7 shows that a nonzero odd realization can arise only by adding a nonzero Γ-odd kernel element of the canonical equation.

Theorem 6.9 (Raw descendant data for a chosen monotone)**.** *Assume assumption 1.1. Fix a finite-cutoff UOM receiver monotone $M(u)$ in the anomaly/Wess–Zumino sector, realized by compact head smearing (1), and assume $M(u_0) \neq 0$. Then at finite cutoff Λ_k, on the slab-supported sufficiently local monotone sector, the following hold:*

1. *a geometric bicomplex $(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k), Q_t, \mathfrak{D}_u)$ with*

$$Q_t^2 = \mathfrak{D}_u^2 = Q_t\mathfrak{D}_u + \mathfrak{D}_u Q_t = 0;$$

2. *a fixed-scheme radial Ward decomposition on the monotone sector,*

$$Q_u = Q^{\mathrm{phys}}_u + Q^{\mathrm{scheme}}_u, \qquad Q^{\mathrm{scheme}}_u = 0;$$

3. *on compactly smeared monotone head observables,*

$$Q^{\mathrm{phys}}_u X = -\mathfrak{D}_u X;$$

4. *the t-Ward identity $\langle Q_t X\rangle_\rho = 0$ on the physical SK sector;*

5. *a good monotone representative $\mathcal{O}^{(0)}_k(u) \in \mathcal{F}_{\mathrm{s.loc}}{}^{\Lambda_k}_{\mathrm{mon}}(\mathscr{X}_k)$ with*

$$M(u) = \left\langle \mathcal{O}^{(0)}_k(u)\right\rangle_{\rho_u}, \qquad Q_t\mathcal{O}^{(0)}_k(u) = 0, \qquad [\mathcal{O}^{(0)}_k(u_0)] \neq 0 \in H^0(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k), Q_t);$$

6. *the monotone derivative identity*

$$\partial_u M(u) = -\left\langle Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) \right\rangle_{\rho_u};$$

7. *the canonical sufficiently local anomaly insertion* $\mathcal{A}_k^{(1)}(u) := Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u)$ *and a raw first-descendant potential* $\Xi_k^{(1)}(u) := 0$ *such that*

$$Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) + Q_t \Xi_k^{(1)}(u) = \mathcal{A}_k^{(1)}(u), \qquad Q_t \mathcal{A}_k^{(1)}(u) = 0;$$

8. *the exact descendants for the canonical anomaly representative are exactly the vector space*

$$\ker(Q_t) \cap (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}};$$

9. *if* $\partial_u M(u_0) \neq 0$, *then*

$$[Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u_0)] \neq 0 \quad \textit{in} \quad H^1(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k), Q_t);$$

10. *the canonical core cochains are swap-even:*

$$\Gamma \mathcal{O}_k^{(0)}(u) = \mathcal{O}_k^{(0)}(u), \qquad \Gamma \mathcal{A}_k^{(1)}(u) = \mathcal{A}_k^{(1)}(u).$$

Proof. Item (1) is lemma 2.3. Item (2) is proposition 3.3. Item (3) is proposition 3.4. Item (4) is proposition 2.5. Item (5) is theorem 3.7. Item (6) is lemma 4.2. Item (7) is theorem 4.5. Item (8) is proposition 4.7. Item (9) is proposition 4.8. Item (10) is proposition 6.4. □

Theorem 6.10 (Conditional Hodge-normalized extension)**.** *Assume the hypotheses of theorem 6.9 and assumption 5.4. Then, for the same chosen monotone* $M(u)$, *every admissible odd representative* $\widetilde{\mathcal{A}^{(1)}}_k(u) \in \mathcal{C}_{\mathrm{adm},k}(u)$ *of the radial anomaly class* $[\mathcal{A}_k^{(1)}(u)]$ *defines a representative defect*

$$\widetilde{B}_k(u) := \widetilde{\mathcal{A}^{(1)}}_k(u) - \mathcal{A}_k^{(1)}(u) \in \mathrm{ran}(\overline{Q_t}_{\mathrm{adm}})$$

which admits the canonical Hodge-normalized primitive

$$\widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u) = \overline{Q_t}_{\mathrm{adm}}^{\dagger} \big(\overline{Q_t}_{\mathrm{adm}}\, \overline{Q_t}_{\mathrm{adm}}^{\dagger}\big)^{-1} \widetilde{B}_k(u) \in \mathcal{H}_{\bar{0},\mathrm{adm}}^{\Lambda_k}(\mathscr{X}_k; u).$$

This primitive satisfies

$$Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) + \overline{Q_t}_{\mathrm{adm}}\, \widetilde{\Xi^{(1)}}_{\mathrm{can},k}(u) = \widetilde{\mathcal{A}^{(1)}}_k(u)$$

in the admissible closed Hilbert sector.

Proof. This is theorem 5.7. □

Theorem 6.11 (Exact swap-odd extension)**.** *Assume the hypotheses of theorem 6.9. Then every raw solution* $\Xi_k^{(1)}(u)$ *of the exact anomalous descent equation yields a* Γ*-odd descendant candidate*

$$\Xi_{-,k}^{(1)}(u) = \frac{1}{2}\big(\Xi_k^{(1)}(u) - \Gamma \Xi_k^{(1)}(u)\big)$$

solving the same exact anomalous descent equation. For the canonical anomaly representative $\mathcal{A}_k^{(1)}(u) = Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u)$, *the* Γ*-odd exact descendants are precisely*

$$\ker(Q_t) \cap (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}} \cap \ker(\Gamma + \mathrm{id}).$$

Proof. The first statement is corollary 6.6. The second statement is corollary 6.7. □

7 Finite-Cutoff Realization of $2\mathrm{T}_{\mathrm{RG}}$

Definition 7.1 (Imported UV-local sufficiently local class)**.** Fix a double cone O in the Minkowski UV chart with $O \subset \mathscr{X}_k$. Let

$$\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(O) \subset \mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k)$$

denote the UOM-P5 local subspace of sufficiently local gauge-invariant functionals supported in O.

Proposition 7.2 (Double-cone localization of the cumulative RG data)**.** *Assume assumption 1.1. Fix a double cone $O \subset \mathscr{X}_k$ in the Minkowski UV chart and choose the compact kernels in (1) with support contained in O. Then*

$$\mathcal{O}^{(0)}_k(u) \in \mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(O), \qquad \mathcal{A}^{(1)}_k(u) \in \mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(O), \qquad \Xi^{(1)}_{\mathrm{sel},k}(u) := \mathcal{O}^{(0)}_k(u) \in \mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(O).$$

Moreover, on the canonical monotone sector in the chosen fixed scheme, the derivations Q_t, Q^{phys}_u, and Q^{scheme}_u preserve $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(O)$.

Proof. By Proposition 4.11, compactly smeared receiver-head scalar densities belong to $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k)$; the chosen support condition then places $\mathcal{O}^{(0)}_k(u)$ in $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(O)$. The selected descendant potential $\Xi^{(1)}_{\mathrm{sel},k}(u)$ is $\mathcal{O}^{(0)}_k(u)$ itself, hence also belongs to $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(O)$.

By proposition 3.4, on compactly smeared monotone head observables one has $Q^{\mathrm{phys}}_u = -\mathfrak{D}_u$. The geometric derivative $\mathfrak{D}_u$ acts only on the local density and the fixed compact kernels in (1); it does not enlarge support. Therefore $\mathcal{A}^{(1)}_k(u) = Q^{\mathrm{phys}}_u \mathcal{O}^{(0)}_k(u) \in \mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(O)$. By proposition 3.3, $Q^{\mathrm{scheme}}_u = 0$ in the chosen fixed scheme, so it trivially preserves $\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(O)$. The same support-preservation argument applies to Q_t, since it acts by Lie differentiation in t on the full smeared density without enlarging support. □

Definition 7.3 (Cumulative anomaly-floor monotone candidate)**.** Assume assumption 1.1 and fix a branch interval

$$(u_0 - \varepsilon, u_{\mathrm{IR}}]$$

with $u_{\mathrm{IR}} > u_0$. Let $L_{\mathrm{af}}(t, x; u)$ be a receiver-head scalar density in the anomaly/Wess–Zumino sector whose compact head smearing yields the finite-cutoff anomaly-floor observable appearing in (42) and (185). Define the cumulative density

$$L_{\mathrm{cum}}(t, x; u) := \int_u^{u_{\mathrm{IR}}} L_{\mathrm{af}}(t, x; v)\, dv$$

and choose the monotone density in definition 3.1 to be

$$L_{\mathrm{mon}} := L_{\mathrm{cum}}.$$

Write

$$\mathcal{O}^{\mathrm{af}}_k(u) := \int_{\mathscr{X}_k} dt\, d^3x\, \kappa_k(t)\eta_k(x) L_{\mathrm{af}}(t, x; u), \qquad \mathcal{O}^{(0)}_k(u) := \int_{\mathscr{X}_k} dt\, d^3x\, \kappa_k(t)\eta_k(x) L_{\mathrm{cum}}(t, x; u).$$

Then

$$\mathcal{O}^{(0)}_k(u) = \int_u^{u_{\mathrm{IR}}} \mathcal{O}^{\mathrm{af}}_k(v)\, dv.$$

Proposition 7.4 (Activity of the cumulative anomaly-floor monotone)**.** *Assume assumption 1.1 and definition 7.3. Let*

$$\Theta(u) := \left\langle \mathcal{O}_k^{\mathrm{af}}(u) \right\rangle_{\rho_u}, \qquad M(u) := \left\langle \mathcal{O}_k^{(0)}(u) \right\rangle_{\rho_u} = \int_u^{u_{\mathrm{IR}}} \Theta(v)\, dv.$$

Then at every active branch point with nonzero admitted payload,

$$\Theta(u) > 0, \qquad M(u) > 0, \qquad \partial_u M(u) = -\Theta(u) < 0.$$

In particular, for every active UV-side point u_0,

$$M(u_0) \neq 0, \qquad \partial_u M(u_0) < 0.$$

Proof. By (126) and Proposition 4.2, the admitted payload is a nonnegative classical accepted-record quantity. On the welded band-locked locus, (185) gives

$$\Theta(u) = K(\Lambda(u))\, \Delta I(u),$$

with $K(\Lambda) > 0$. Therefore every active tick with nonzero admitted payload satisfies $\Theta(u) > 0$.

By definition 7.3,

$$M(u) = \int_u^{u_{\mathrm{IR}}} \Theta(v)\, dv,$$

so positivity of $\Theta(v)$ on the active branch gives $M(u) > 0$. Differentiating with respect to the lower limit yields

$$\partial_u M(u) = -\Theta(u) < 0.$$

Evaluating at u_0 gives the final displayed statement. □

Lemma 7.5 (Explicit selector from the cumulative anomaly-floor monotone)**.** *Assume assumption 1.1 and definition 7.3. Define the explicit raw first-descendant selector by*

$$\Xi_{\mathrm{sel},k}^{(1)}(u) := \mathcal{O}_k^{(0)}(u).$$

Then:

1. $\Xi_{\mathrm{sel},k}^{(1)}(u) \in (\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}}$ *and* $Q_t \Xi_{\mathrm{sel},k}^{(1)}(u) = 0$*;*
2. *the anomaly insertion*
$$\mathcal{A}_k^{(1)}(u) := Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) = \mathcal{O}_k^{\mathrm{af}}(u)$$
is sufficiently local, Q_t-closed, and satisfies the exact anomalous descent equation
$$Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u) + Q_t \Xi_{\mathrm{sel},k}^{(1)}(u) = \mathcal{A}_k^{(1)}(u);$$
3. *if u_0 is an active UV-side point, then $\Xi_{\mathrm{sel},k}^{(1)}(u_0)$ is not a c-number on the connected vacuum component of the reduced phase space.*

Proof. Because $L_{\mathrm{cum}}(t, x; u)$ is a receiver-head scalar density obtained by integrating the local anomaly-floor density $L_{\mathrm{af}}(t, x; v)$ over the external branch parameter v, the compact smearing $\mathcal{O}_k^{(0)}(u)$ belongs to $(\mathcal{F}_{\mathrm{s.loc}}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}}$. Its Q_t-closure is exactly proposition 3.2, proving item (1).

For item (2), proposition 3.4 gives $Q_u^{\rm phys} = -\mathfrak{D}_u$ on compact head smearings. Since

$$\mathfrak{D}_u L_{\rm cum}(t,x;u) = -L_{\rm af}(t,x;u),$$

one has

$$Q_u^{\rm phys}\mathcal{O}_k^{(0)}(u) = -\,\mathfrak{D}_u\mathcal{O}_k^{(0)}(u) = \int_{\mathscr{X}_k} dt\, d^3x\ \kappa_k(t)\eta_k(x)L_{\rm af}(t,x;u) = \mathcal{O}_k^{\rm af}(u).$$

Thus $\mathcal{A}_k^{(1)}(u) = \mathcal{O}_k^{\rm af}(u)$. The sufficient locality and Q_t-closure of $\mathcal{A}_k^{(1)}(u)$ are the same arguments as in theorem 4.3 and proposition 4.4. Since $Q_t\Xi_{{\rm sel},k}^{(1)}(u) = 0$, the displayed exact anomalous descent equation follows.

For item (3), suppose instead that $\Xi_{{\rm sel},k}^{(1)}(u_0) = \mathcal{O}_k^{(0)}(u_0)$ were constant on the connected vacuum component of the reduced phase space. Choose a nonzero compact welded trace perturbation τ supported where the compact kernels are nontrivial, and let $\varphi_{u_0,s} := \varphi_{u_0} + s\tau$ be the corresponding local line in the same connected vacuum component for small s. By (41) and (42), the instantaneous anomaly floor is a positive quadratic form of the welded trace perturbation. Therefore, at fixed u_0,

$$\mathcal{O}_k^{(0)}(u_0)[\varphi_{u_0,s}] = \mathcal{O}_k^{(0)}(u_0)[\varphi_{u_0,0}] + s^2\, C_{u_0}(\tau) + o(s^2)$$

for some $C_{u_0}(\tau) > 0$. Hence $\mathcal{O}_k^{(0)}(u_0)$ is not constant on that connected vacuum component, a contradiction. So $\Xi_{{\rm sel},k}^{(1)}(u_0)$ is not a c-number there. □

Theorem 7.6 (Finite-cutoff realization of $\mathbf{2T}_{\rm RG}$)**.** *Assume assumption 1.1 and definition 7.3. Choose the monotone representative of definition 3.1 from the cumulative anomaly-floor density $L_{\rm mon} = L_{\rm cum}$, and let*

$$M(u) := \left\langle \mathcal{O}_k^{(0)}(u)\right\rangle_{\rho_u}.$$

Then, at finite cutoff Λ_k and on the slab-supported sufficiently local monotone sector, the following $\mathbf{2T}_{\rm RG}$ data are realized:

1. *a sufficiently local good monotone representative $\mathcal{O}_k^{(0)}(u)$ such that*

$$M(u) = \left\langle \mathcal{O}_k^{(0)}(u)\right\rangle_{\rho_u}, \qquad Q_t\mathcal{O}_k^{(0)}(u) = 0, \qquad [\mathcal{O}_k^{(0)}(u_0)] \neq 0 \in H^0(\mathcal{F}_{\rm s.loc}^{\Lambda_k}(\mathscr{X}_k), Q_t);$$

2. *fixed-scheme RG Ward data and scheme control on the monotone sector,*

$$Q_u = Q_u^{\rm phys} + Q_u^{\rm scheme}, \qquad Q_u^{\rm scheme} = 0, \qquad Q_u^{\rm phys}X = -\mathfrak{D}_u X$$

on compactly smeared monotone head observables, together with

$$\partial_u M(u) = -\left\langle Q_u^{\rm phys}\mathcal{O}_k^{(0)}(u)\right\rangle_{\rho_u}, \qquad \left\langle Q_u^{\rm scheme}\mathcal{O}_k^{(0)}(u)\right\rangle_{\rho_u} = 0;$$

3. *explicit sufficiently local anomalous first descent,*

$$\mathcal{A}_k^{(1)}(u) := Q_u^{\rm phys}\mathcal{O}_k^{(0)}(u) = \mathcal{O}_k^{\rm af}(u) \in (\mathcal{F}_{\rm s.loc}^{\Lambda_k}(\mathscr{X}_k))_{\bar{1}}, \qquad \Xi_{{\rm sel},k}^{(1)}(u) := \mathcal{O}_k^{(0)}(u) \in (\mathcal{F}_{\rm s.loc}^{\Lambda_k}(\mathscr{X}_k))_{\bar{0}},$$

with

$$Q_u^{\rm phys}\mathcal{O}_k^{(0)}(u) + Q_t\Xi_{{\rm sel},k}^{(1)}(u) = \mathcal{A}_k^{(1)}(u), \qquad Q_t\mathcal{A}_k^{(1)}(u) = 0,$$

and the chosen UV-side selector $\Xi_{{\rm sel},k}^{(1)}(u_0)$ is not a c-number on the connected vacuum component;

4. *UV-local double-cone localization of the cumulative canonical data in the sense of proposition 7.2;*

5. *a scheme-stable degree-one anomaly class: for every cohomologically admissible finite-cutoff inter-scheme comparison, the class represented by* $\mathcal{A}_k^{(1)}(u)$ *is invariant in* $H^1(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k), Q_t)$ *and in the local ambiguity quotient;*

6. *the cumulative anomaly-floor monotone has nontrivial UV slope*

$$\partial_u M(u_0) < 0$$

at every active UV-side point u_0 *on the chosen branch.*

Consequently, if u_0 *is such an active UV-side point, then*

$$[Q_u^{\mathrm{phys}} \mathcal{O}_k^{(0)}(u_0)] \neq 0 \qquad \textit{in} \qquad H^1(\mathcal{F}^{\Lambda_k}_{\mathrm{s.loc}}(\mathscr{X}_k), Q_t).$$

Thus the slab-supported monotone sector carries an explicit finite-cutoff realization of $\mathbf{2T}_{\mathrm{RG}}$*, with fixed-scheme representatives and a scheme-stable degree-one anomaly class.*

Proof. Item (1) is theorem 3.7 together with proposition 7.4. Item (2) is the combination of propositions 3.3 and 3.4 and lemma 4.2. Item (3) is lemma 7.5. Item (4) is proposition 7.2. Item (5) is theorem 4.10. Item (6) is proposition 7.4. The final nontriviality statement is then proposition 4.8. □

Remark 7.7 (Later selector, spectral, and transport uses). The raw swap-odd class of theorem 6.11 and the explicit $\mathbf{2T}_{\mathrm{RG}}$ data of theorem 7.6 are the UOM-side inputs used later by theorems 1.9, 3.19 and 4.45.